# Changelog

## 1.8.1

Two regressions found by stress-testing 1.7.0/1.8.0 rather than assuming they were sound. Both were introduced by the optimization work itself.

### Fixed - `run_batch` could fire a destructive operation during collection

`run_batch` used a DENYLIST of non-batchable tools, so anything not listed ran for real during what should have been a dry collection pass. Verified: batching `clear_document` actually WIPED the document and then reported the batch as failed - data loss plus a misleading error.

Now fail-safe: - An explicit **allowlist** (27 geometry/parameter tools). A tool omitted simply cannot be batched, instead of silently executing. - Every operation is validated **before** collection mode is entered, so an invalid batch has no side effect at all. - `_send_async` refuses outright while collecting, so the async path cannot fire. - A reentrant lock around collection, so a concurrent call cannot be swallowed into the batch buffer and silently never executed.

### Fixed - terse mode hid the body of report tools

The 1.8.0 compaction dropped indented continuation lines, but the entire output of `document_report` is indented - so the tool built specifically to prevent wrong-document overwrites was reduced to its first line. `get_state`'s open-document list, `list_faces`, `cross_section` boundaries and others were affected the same way.

Compaction is now opt-in per call: 18 report tools pass `compact=False`, because their body IS the detail. Action tools still compact on clean success, so the token saving is retained where it was actually free.

## 1.8.0

The remaining measured optimizations from the 1.7.0 profiling pass.

### Claude tokens

- **Terse-on-success output ( `set_verbosity` ).** Detail is valuable exactly when something is WRONG; on a clean success it is pure token cost paid on every call. Indented explanatory lines are now hidden when the payload contains no trouble marker (***, WARNING, ERROR, FAILED, UNCONFIRMED,

MISMATCH, REFUSED, Traceback, no-op, STALE). Any one of those disables compaction for the whole payload, so failures always print in full. `set_verbosity('full')` restores everything.

### Host compute

- **Face-iteration caps.** `get_model_state` computed the area of every face of every body to count slivers, and `list_faces` measured every face before capping the output - each area/centroid read is a kernel query. Both now bound at 4000 faces and say so, instead of stalling on a dense model.
- `cross_section` **refines boundaries by bisection** instead of scanning uniformly. A uniform scan wastes queries deep inside solid regions and still only resolves boundaries to span/N. Now: coarse pass, then ~8 bisections per transition. Measured on a hollow box with walls at 10.700/49.300 mm: reported 10.699/49.301 mm from 47 kernel queries, versus 1.94 mm precision from the old 31-query uniform scan - roughly 1900x better precision for 16 extra queries. Also reports span widths and the query count.

### Round trips

- `job_status` **documents polling backoff** (~5s, 10s, 20s, then 30s). Each poll is a round trip that returns the whole accumulated stdout again; get_state is cheaper when only liveness is needed.

## 1.7.0

Optimization pass - measured, not guessed.

### Host CPU (a regression introduced in 1.6.0)

- **The identity refresh was a 4-second heartbeat.** It queued a Fusion main-thread job every 4 seconds for as long as Fusion was open (~21,600 interrupts/day), each enumerating every open document and touching dataFile - which raises, expensively, on builds where app.data is broken. Cost was paid continuously even when nobody was using the tool. It is now LAZY: /state requests a refresh only when its cache is actually stale, coalesced to at most one in flight and one per 2 s. Idle cost is now zero.

### Speed - round trips

- `run_batch(operations_json)` fuses N tool calls into ONE round trip and ONE Fusion job. Every tool call otherwise costs a full hop (model -> server -> HTTP -> main thread -> back), so ten holes cost ten hops. Each operation still prints its own output and per-step volume deltas; stop_on_error aborts at the first failure. Verified: a 5-operation batch used 1 round trip instead of 5. Not for steps whose arguments depend on an earlier step's result.

### Claude tokens

- **Tool schemas are sent on EVERY request.** Measured at ~5,488 tokens. Seven of the largest docstrings were trimmed to their imperative warnings, with the reference material moved to a new on-demand `fusion_help(topic)` tool (topics: helpers, planes, traps, workflow). Now ~4,677 tokens - a saving of **~811 tokens on every request**, roughly 40k over a 50-turn session.
- `view_model` **resolution is now a parameter, defaulting to 800x600 instead of 1200x900.** An image costs about (w*h)/750 tokens, so the default drops from ~1,440 to ~640 - ~800 tokens saved per screenshot*, the single biggest per-call saving. Raise it only when judging fine detail; 640x480 (~410 tokens) suffices for a quick sanity look.

## 1.6.1

Audit for other instances of the bug class behind 1.6.0 (implicit target / unverified success). Found four more, all real, all fixed.

- **40 tools assumed** `app.activeProduct` **IS a Design.** It is not, in the Drawing, CAM, Render or Simulation workspace - `.rootComponent` then fails obscurely or targets something unexpected. All 40 now route through an injected `_require_design()` that raises a clear error naming the actual product type and document, and telling the user to switch workspace.
- **Exports announced success without checking.** `em.execute()` 's return value was discarded and the tool printed "exported ..." regardless - the same "thinks it saved but it didn't" failure applied to files. STEP/STL/DXF now verify the return value AND that the file exists with non-zero size, printing CONFIRMED with the byte count or an explicit failure.
- `set_parameter` **never read back.** Fusion silently rejects or clamps an expression it does not accept, so a parameter could report as set to 50 mm while holding something else. It now reads the stored expression back and reports CONFIRMED or MISMATCH with both values.
- **Joints reported success without verifying.** `joints.add()` returning without error does not mean a joint exists. Now checks the joint count increased.
- `clear_document` **- the most destructive tool - checked nothing.** It now prints the target document name, lineage, and current contents before deleting anything.

## 1.6.0

Document identity. Claude could not reliably tell which document was open, which produced the worst failures in the project: overwriting an unrelated design, starting a new document instead of continuing in the open one, and reporting a save that never happened. Four compounding root causes, all fixed.

- `doc.dataFile` **routes through** `app.data` **.** When app.data is dead (common on this build, survives a Fusion restart), lineage read as missing - and the old logic treated "no lineage" as "never saved", so it called `saveAs` and created a DUPLICATE of a document that was already saved. `save_document` now tries the NON-DESTRUCTIVE `doc.save()` first whenever lineage is unreadable, and refuses to

fall back to `saveAs` without explicit `force_new_lineage=True`. Verified: in that scenario it now makes zero saveAs calls.

- **Identity was a stale cache.** `get_state` only refreshed after a job ran, so switching documents in the Fusion UI left it reporting the previous one. The listener now refreshes identity on a timer, and `/state` reports its own snapshot age and a `fresh` flag.
- `isSaved` **was the sole branch signal.** Every identity signal (`isSaved`, `isModified`, `dataFile`, version, folder) is now read INDEPENDENTLY and reported side by side, so a contradiction is visible instead of collapsed.
- **No visibility of open documents.** `document_report()` is a live, authoritative read listing EVERY open document and which is active, ending in an explicit verdict that distinguishes "never saved" from "cannot tell". `activate_document(name_or_index)` binds work to the user's document.
- **Saves are now verified, not assumed.** `save_document` reports version before/after and prints CONFIRMED only when an independent signal agrees (version bump / modified cleared / lineage appeared); otherwise UNCONFIRMED, with instructions not to report the document as saved.

## 1.5.0

Round-2 field report (organic modelling: spline surfaces of revolution, lofted bifurcation, swept elbows, ~15 fillet campaigns).

### Added - edge selection & fillet verification (the biggest gap)

- `select_edges(...)` - predicate selection by adjacent **surface-type pair**, z range, and length, returning tempIds. `which=all/vertical/top` could not express what every real fillet needs, so the good search logic in `fillet_edges_smart` was locked behind a selector that couldn't reach it - the same ~30-line predicate was hand-rolled about fifteen times in one session.
- `list_edges` **now reports the surface-type pair** (e.g. `Cylinder|Cone`), so callers stop re-implementing the classifier.
- `fillet_edges_smart` **accepts** `edge_tempids` and now verifies **topology as well as volume**: it reports which faces disappeared and warns when a face smaller than `4r^2` was consumed. Round 1's failure was a fillet that ADDED geometry far away; this session's was the inverse - a 2.0 mm fillet that ate a 1.3 mm rim annulus and rounded the tube ends away while the volume delta (-18 mm^3 vs ~28 expected) passed every threshold.
- `fillet_preview(...)` - shows the affected faces and flags any smaller than `4r^2` BEFORE committing.
- **A zero volume delta is now reported as a RESULT, not a failure** - it means the surfaces already meet tangentially and need no fillet.

## Added - escape hatches

- `force_delete_feature(name)` - resolves a MUTUAL-REFERENCE DEADLOCK where two features each blame the other and neither suppress nor `deleteMe()` works in any order. Rolls the marker back before deleting (what the Fusion UI does), then reports what else broke. This turned a blocked session into a 2-minute fix.
- `sweep_tube_check(...)` - measures a path's minimum curvature radius against the tube radius. `ASM_SELF_INTER` blames the profile/taper, but on a circular profile the PATH is the cause.
- `respline(...)` - rebuild a fitted spline with new fit points (Fusion can move but not add them). States honestly that downstream references break, and reports which features became unhealthy.

## Fixed

- **Degenerate-edge crash.** Tangentially-meeting lofts leave zero-length edges with <2 faces; `e.faces.item(1)` raised `Bad index parameter`. Guards are now inside the listener helpers (`edge_ok`, `safe_edges`, `edge_face_pair`) and every edge-walking tool, so callers stop rediscovering it via traceback.
- `explain_fillet_error` **knew almost none of the codes.** Added `ASM_ABL_NO_GEOM_EF`, `ASM_ABL_CVXTY_VIOLATION`, `ASM_BL_NO_VTX_GEOM`, `ASM_BL_CHG_PT_COMPLEX_VERT`, `ASM_BL_CANNOT_REORDER`, `ASM_INCONS_VERT_ATT`, `ASM_SELF_INTER`, `FILLET_NO_EDGE_FOUND`, `EDGE_REFERENCE_LOST`.
- `refresh_geometry()` - `computeAll()` does not guarantee a fresh `body.volume`; identical readings after a sketch edit nearly led to concluding a correct edit had failed.
- `path_report(path)` - segment count + length, so a mis-chained path (which silently produces a short sweep) is visible.

## Notes on "did not land" items

- **clear_document (item 5):** real, and fixed in 1.4.1 - the 1.4.0 rewrite was correct but kept a fallback that still bulk-deleted bodies, the exact hang path.
- **YZ docstring (item 7):** corrected in 1.4.0. The docstring now carries its own version stamp, because Claude Desktop caches tool schemas at startup - a fixed docstring can still *display* stale without a full restart.
- **app.data (item 6):** `save_document`'s fallback works and is verified. `open_document` / `list_recent_designs` genuinely cannot be worked around; this is now stated plainly rather than implied.
- `ping_fusion` **reports both server and listener versions** and warns when they differ (added 1.4.1) - a stale *server* previously produced no mismatch signal.

## 1.4.1

Follow-up to two field reports that turned out to be **version mismatch** - both symptoms were v1.2.1 behaviour, and both were already fixed in 1.4.0. The real finding was that nothing made that diagnosable, plus one genuine remaining hazard.

- `clear_document` **fallback sweep hardened.** The 1.4.0 rewrite used reverse-order `entity.deleteMe()` correctly, but kept a fallback that still bulk-deleted `bRepBodies` - the exact path that hung for 10+ minutes on a parametric document. That sweep now runs ONLY when there is no timeline (direct-modelling docs). On a parametric doc with leftovers it reports what remains and points at `retire_body()` instead of grinding. Verified: zero bulk deletes on a parametric doc.
- `ping_fusion` **now reports BOTH server and listener versions** and warns when they differ. The existing protocol handshake only caught a stale *listener*; a stale *server* (this case) produced no mismatch signal at all, so an old docstring and an old code path looked like live bugs.

## 1.4.0

Fixes from a full modelling session (splitter funnel). Several of these were silent-wrong-geometry bugs, which are the expensive kind.

### Fixed (P0 - wrong facts that produced wrong geometry)

- **The documented YZ sketch-plane mapping was WRONG.** Docs claimed `sketch X -> world Y, sketch Y -> world Z`; measured behaviour is `sketch X -> world -Z, sketch Y -> world +Y`. Building off the wrong mapping yields a body with a correct bounding box and correct profile area that only reveals itself several operations later. Docstring corrected, and new injected helpers `P_xy/P_xz/P_yz` (world mm in, correct sketch point out) plus `verify_plane_mapping(sketch)` mean nobody has to reason about it.
- `report_volume_delta` **cried wolf on every Join.** It summed ALL bodies, so a Join that merges two bodies is volume-neutral and got flagged as a no-op - training you to ignore the one warning that matters. Now takes a `capture_state(target_body)` snapshot and warns only when body count AND volume are both unchanged.
- **Nothing detected silent geometry destruction by fillets.** A tangent-chained 1.5 mm fillet propagated ~20 mm and reverted a bore radius, reporting success with a valid body and no slivers. `fillet_edges` now checks the volume delta against `~0.1*r^2*L` and warns on an order-of-magnitude overshoot; `fillet_volume_sanity()` exposes the same check to raw code.

### Fixed (P1 - broken tools)

- `export_stl` **was unusable at every quality setting.** The enum names `*QualityTriangleMeshRefinement` do not exist on this build; the real members are

`MeshRefinementLow/Medium/High/Custom`. Now uses those, with a graceful fallback that prints the available members if they change again.

- **`app.data` failures killed save/list/open for a whole session.** `save_document` now detects `InternalValidationError`, falls back to `doc.save()` for an already-saved document (the path that does work), and says clearly that a new lineage needs the Fusion UI. `list_recent_designs` explains the failure instead of returning a bare error.
- **`clear_document` hung for 10+ minutes and took Fusion down.** Reimplemented as reverse-order `item(i).entity.deleteMe()` (TimelineObject has no `deleteMe`) with `progress()` per feature, and it now runs through the async job system by default so the channel is never blocked.

## Fixed (P2)

- **Bodies added via `BaseFeature` return a stale reference** - `ALL_TOOL_BODY_REFERENCE_LOST` when reused as a Combine tool. `cut_prisms` and `commit_tool_body()` now re-fetch by name after `finishEdit()`. This was a real bug in code shipped in 1.3.0.

## Improved (P3 - observability)

- **`view_model`**: added `bottom`; an unknown orientation is now an ERROR instead of silently returning the previous view; added `zoom_x/y/z_mm` + `zoom_radius_mm` to frame a small detail instead of always fitting the whole model.
- **`cross_section`**: added `at_a_mm` / `at_b_mm` to pin the held axes, so you can scan through a specific feature rather than the bbox mid-plane.
- **`fillet_edges_smart`** - searches tangent-chain on/off x descending radii, verifies volume delta, rolls back overshoots, reports what worked. **`explain_fillet_error`** translates raw ACIS/ASM codes into fixes.
- **`smallest_faces(body)`** prints the actual smallest face areas; `sliver_face_count` threshold raised to 0.05 mm^2 (0.010-0.014 mm^2 faces are visible artifacts that the old cutoff missed).
- **Transport timeouts** now return a structured RUNNING status naming the job id, with an explicit "do not re-issue" note - a timed-out call had already completed a save in one case.
- **`retire_body()`** + docs for the replace-a-body idiom (`removeFeatures.add(old)`), which is strictly safer than clearing.

## Docs

- New "API traps on this build" section in TOOLS.md covering the plane mapping table, `TimelineObject.entity`, stale BaseFeature handles, `is`-comparison on sketch entities, `addByCenterStartSweep` normalisation, `setMidRadii` list types, `app.data` failures, and silent fillet propagation.

## 1.3.0

Security fix plus the curved-surface toolset the docs already promised.

### Fixed (P0)

- **SECURITY: argument injection.** Tool arguments were pasted directly into Python source text, so a filepath or name containing a quote broke out of the literal and executed arbitrary code inside Fusion. All ~22 interpolation sites now emit proper Python literals via `repr()`. Verified by executing generated code for 9 tools x 7 adversarial payloads (63 runs) with an injection canary: zero escapes.
- **P0: Windows paths raised SyntaxError.** Same root cause - `C:\parts\x.step` was pasted raw, so backslashes became escape sequences. Fixed by the same change; paths with backslashes, trailing separators and apostrophes now work.

### Added (P1)

- `wrap_cut` - wrap a flat sketch onto a non-planar face and cut/emboss to depth. Previously referenced by the docs but MISSING from the build, so following the instructions led straight into a wall. Uses the correct `EmbossFeatures.createInput(profiles, faces, depth)` signature (negative depth cuts), cuts per-profile with individual volume checks, and reports ok/no-op/failed counts with sector guidance.
- `project_to_surface`, `construction_plane_offset`, `construction_plane_tangent`, `sketch_axis_map` - the supporting cast the wrap workflow needs. The construction-plane tools replace `ConstructionPlaneInput.setByPlane`, which is environment-blocked here, and echo world position so unit slips surface immediately.
- `cut_prisms` + injected `tbm()` / `commit_tool_body()` helpers - many prism cuts in ONE timeline feature via `TemporaryBRepManager`. Depends on neither construction planes nor face tempIds, so it sidesteps the two most fragile things in this environment.

### Docs

- TOOLS.md gains a curved-surface section and a TemporaryBRepManager section with a worked example.

## 1.2.1

Fixes from a project-wide review triggered by a real degraded-state incident (old listener + new server = every payload empty while status read OK).

- **CRITICAL: switched to ThreadingHTTPServer.** The v1.2.0 listener used a single-threaded HTTPServer, so while `/exec_sync` held the one HTTP thread waiting on a job (up to ~110s), `/ping` and `/state` could not answer - silently defeating the "separate always-responsive channel" design. Verified fixed: ping now answers instantly during a blocked sync call.

- **Version/protocol handshake.** The listener now reports `listener_version` + `protocol` in every response; the server checks it and returns a loud PROTOCOL MISMATCH message with exact fix steps when the add-in is stale - instead of silently rendering empty payloads as STATUS: OK. This makes the incident that prompted this release impossible to miss.
- **Fail-loud capture health.** Envelopes include `stdout_bytes`; the server marks ok-but-silent results with "(no output captured)".
- **Job-store pruning.** Finished jobs are capped at the 50 most recent so long sessions don't grow memory unboundedly.
- **Docs:** INSTALL now warns that Fusion loads the add-in from its AddIns folder, not from wherever the project files live - updating one does not update the other.

## 1.2.0

Async execution and document safety — driven by a second round of real-use feedback. The core fix: a 4-minute timeout that told you nothing about what actually happened in Fusion. Execution is now decoupled from the HTTP response.

### Added — async job model (the big one)

- **Listener is now async.** `POST /exec` enqueues a job and returns a `job_id` immediately; the operation runs on the main thread and is polled via `GET /job/<id>`, which returns a structured envelope (`status`, `stdout`, `traceback`, `elapsed_ms`) and **partial stdout while still running** (incremental progress).
- **`run_async(code)`** + **`job_status(job_id)`** tools for long operations that shouldn't block. Plus a `progress(msg)` helper to mark stages visible mid-run.
- **Ping/state on a separate channel.** `GET /ping` and `GET /state` are served from cache without the main thread, so they STAY RESPONSIVE during (or after) a hung operation. A hung op and a dead listener are now distinguishable.
- **`get_state()`** — cheap, guaranteed-fast document identity + counts (no geometry work). The safe first call for crash recovery.
- **Structured result envelope, always.** Every execution returns status/stdout/traceback/elapsed; no more bare "no result received". The sync path returns partial output and a job id if the op outlives the wait.
- **Watchdog** flags jobs that exceed their budget as `timeout` (clear signal rather than a dead channel).

### Added — document safety

- **Document identity surfaced everywhere.** `get_state` and `get_model_state` lead with the active document **name + lineage id**, preventing wrong-document overwrites.

- `save_document(name, create_new=False)` now reports **which lineage it versioned**, and `create_new=True` saves a brand-new lineage so a new part never buries an unrelated design's history.
- `list_versions(design_id="")` — read-only enumeration of a lineage's saved versions, so a bad save is recoverable and the user can be pointed to the exact version to promote.
- `open_document` reports the new active lineage.

### Added — documentation

- Sketch-plane / axis mapping (XY/XZ/YZ + offset planes, incl. the XZ→world-−Z inversion) documented in `run_fusion_code`'s docstring.

### Notes

- `run_fusion_code` uses the bounded sync path (~100s) then hands off to polling; no behavior change for short ops.
- Volume-delta no-op detection and the `mm()` unit helper shipped in 1.1.0 and already cover two earlier feedback items.

## 1.1.0

Observability, diagnostics, and stability. Injected unit/observability helpers (`mm()`, `assert_scale()`, `report_volume_delta()`, etc.); auto volume-delta reporting with no-op warnings on cut/shell tools; `cross_section()`, `list_faces()`, `list_edges()`, `heartbeat()`; listener-level refusal of known host-crashing calls; richer `get_model_state()`.

## 1.0.0

First public release. Fusion 360 add-in listener + MCP server with ~30 tools (primitives, parameters, faces, assemblies/joints, measurement, visual feedback, export, persistence), preflight launcher, and full docs.